

A high-quality genome of the Atacama Desert plant *Cistanthe cachelensis* and its photosynthetic behavior related to drought and life history

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Summary

- Crassulacean acid metabolism (CAM) photosynthesis has independently evolved many times in arid-adapted plant lineages. *Cistanthe cachelensis* (Montiaceae), a desert annual, can upregulate CAM facultatively upon stress such as drought. Few studies, however, consider life history stages when measuring CAM activity or its facultative onset.
- To test the effect of drought and flowering on photosynthetic activity, we assayed *Cistanthe* individuals in fully watered and drought conditions, as well as fully watered individuals at pre-flowering and flowering life stages. We assembled and annotated a chromosome-scale genome of *C. cachelensis* and compared it with the genome of *Portulaca amilis* and analyzed differential gene expression.
- Results show significantly upregulated CAM in drought conditions as compared to fully watered conditions; furthermore, flowering individuals showed slightly higher CAM activity as compared to pre-flowering plants, even when fully watered. Differential gene expression analyses provide preliminary support for the possible coregulation of CAM expression and reproduction.
- We emphasize the potentially missed significance of life history in the CAM literature and consider how the CAM biochemical module could become co-opted into other plant behaviors and responses, such as the shift to reproduction or flowering in annuals.

Introduction

Crassulacean acid metabolism (CAM) photosynthesis has independently evolved many times in diverse lineages, with estimates of up to 7% of all vascular plants having a functional CAM cycle (Gilman *et al.*, 2023). CAM photosynthesis is characterized by the initial nocturnal fixation of CO₂ in the form of a four-carbon acid and its daytime decarboxylation for the final fixation of CO₂ by ribulose-1,5-bisphosphate carboxylase/oxygenase (RuBisCO). This temporal separation of carbon uptake and final fixation allows CAM plants to minimize transpirational water loss, since the stomata are open at night when temperatures are cooler, and thus, evapotranspiration rates are lower (Neales *et al.*, 1968; Osmond, 1978), thus making CAM plants the most water-use efficient plants on Earth (Cockburn, 1985; Nobel, 1996). A major enzyme in this pathway is phosphoenolpyruvate carboxylase (PEPC), which catalyzes the fixation of atmospheric CO₂ into oxaloacetate and then eventually malate, which is stored in

the vacuole overnight as malic acid (Holtum *et al.*, 2005). The fixation of carbon using the CAM cycle does come with energetic costs, with models predicting three times higher nocturnal requirements of adenosine triphosphate (ATP) and reducing agents in CAM compared with C₃, although the costs are mostly alleviated by lower photorespiration rates in CAM plants due to their elevated levels of internal daytime CO₂ concentrations (Shameer *et al.*, 2018).

An important aspect of CAM biology is the diversity of ways in which the biochemical cycle has been integrated into a plant's physiology. In many canonical desert, succulent lineages such as cacti, aloes, and euphorbias, CAM has become the plant's primary photosynthetic metabolism, operating daily and under varied conditions (termed 'strong CAM' in Edwards, 2019; 'primary CAM or pCAM' in Gilman *et al.*, 2024). There are likely even more species, however, that use CAM as an accessory metabolism and still fix the majority of atmospheric CO₂ directly by RuBisCO. Edwards (2019) provided an umbrella term

'C₃ + CAM' to refer to all of these phenotypes ('minority CAM or mCAM' of Gilman *et al.*, 2024), although the phenotypes themselves are quite diverse. They include a weakly expressed constitutive CAM, a fully inducible and reversible facultative CAM, an upregulated CAM cycle that is apparently not reversible, and CAM-cycling/CAM-idling phenotypes, which do not show nocturnal stomatal opening and use the pathway only to refix internal respiratory CO₂ (Winter, 2019).

In plants with a facultative CAM cycle, most experiments focus on environmental factors such as drought, salinity, and occasionally photoperiod as the variables that trigger an upregulation of CAM expression (Winter & Holtum, 2014). Less attention has been paid to endogenous signaling or the incorporation of CAM into particular plant developmental stages or behaviors (Cushman *et al.*, 1990; Herppich *et al.*, 1992; Ting *et al.*, 1996; Adams *et al.*, 1998; Winter & Holtum, 2007; Winter *et al.*, 2011; Ramírez & Herrera, 2017; Winter, 2019). Studies that focus on development as a factor contributing to CAM expression have mostly been in primary CAM species; for example, Winter (2019) shows that nocturnal CO₂ uptake was first measured in *Kalanchoë pinnata* on the 23rd day after leaf emergence. The amount of net nocturnal CO₂ fixation increases over the course of development until the leaf matures to its full succulence (Winter, 2019). Similarly, *Opuntia elatior* facultatively upregulates CAM as a seedling, and primary CAM gradually develops (Winter *et al.*, 2011). On the contrary, the role of development is less clear in the C₃ + CAM *Mesembryanthemum crystallinum*, which necessitates a salt or drought-stress signal for CAM induction (Adams *et al.*, 1998; Winter & Holtum, 2007). Similar investigations in diverse C₃ + CAM lineages are needed to test the role of development and life history in facultative CAM upregulation.

Moreover, there have only been a handful of studies that have investigated possible relationships between CAM expression and reproduction, a very specific ontological event in a plant's development. Plant reproduction represents a substantial carbon investment (Bazzaz *et al.*, 1987), and coupling reproduction with additional nighttime carbon fixation provided by CAM could be advantageous. In one study comparing C₃, C₄, and CAM species, CAM plants had the highest flower biomass and seed-to-fruit biomass ratio (Ramírez & Herrera, 2017). In the C₃ + CAM *M. crystallinum*, individuals under drought and high-salinity conditions achieved 10-fold greater seed production, as compared to individuals grown without these conditions (Winter & Ziegler, 1992). This suggests either that higher CAM activity resulted in higher reproductive success or that when *M. crystallinum* is exposed to stressors it increases its reproductive effort (Winter & Ziegler, 1992; Adams *et al.*, 1998; a behavior also reported in *Talinum triangulare*, Taisma & Herrera, 2003). Furthermore, in perennial CAM plant *Dudleya*, flowering individuals displayed higher rates of nocturnal carbon uptake and nighttime acid accumulation as compared to nonflowering individuals, implying an association between reproduction and CAM (M.M. Bangal & S.C. Lambrecht, unpublished). While each of these studies suggest that CAM might be beneficial in

reproduction, they do not distinguish whether CAM expression is explicitly upregulated with a shift to reproduction or whether its expression as part of a drought response also happens to be beneficial for reproductive output.

Cistanthe cachinalensis is an annual C₃ + CAM species in the Montiaceae (Portulacineae; Caryophyllales; Fig. 1c), native to the Atacama Desert in northern Chile (Fig. 1b). *Cistanthe cachinalensis* flowers during massive 'superblooms' (*desierto florido*) occur in the Atacama Desert, when many thousands of dormant seeds germinate simultaneously – an occurrence usually associated with El Niño years (Vidiella *et al.*, 1999; Jaksic, 2001; Chávez *et al.*, 2019; Fig. 1a). Holtum *et al.* (2021) found that *C. cachinalensis* (referred to as *Cistanthe aff. longiscapa* in that publication) performs low-level constitutive CAM, with the ability to upregulate CAM upon drought. Rewatering did not reverse the drought-induced CAM behavior in *C. cachinalensis* – it maintained an elevated CAM expression throughout the experiment (Holtum *et al.*, 2021). Although not reported, the plant was likely flowering and fruiting during this time, suggesting a possible connection between continued CAM expression and a shift to reproduction.

Importantly, for desert annuals such as *C. cachinalensis* that experience little to no rainfall throughout their development (Eshel *et al.*, 2021; Ossa *et al.*, 2022), drought stress should be most prominent in later stages of life. It is quite likely that, in the field, *C. cachinalensis* is experiencing significant water deficits while flowering; therefore, upregulated CAM expression and reproduction should be temporally correlated during its life history. At the same time, the expression of a CAM cycle during flowering could potentially benefit an annual plant independently of its water-use efficiency, as increased carbon fixation during this time will maximize its resources to allocate to developing seeds (Herrera, 2009; Ramírez & Herrera, 2017). Recent discussions on the evolution of CAM focus less on the assembly of the biochemical module, which is essentially present in all plants, and rather on changes in the regulation of module expression (Bräutigam *et al.*, 2017; Schiller & Bräutigam, 2021; Heyduk *et al.*, 2022; Edwards, 2023). In a similar vein, we wonder whether a 'fully formed' CAM biochemical module that is consistently induced under certain conditions (i.e. drought) could evolve to be expressed in other conditions in which it is advantageous via recruitment into a new regulatory module. Specifically, if CAM expression increases reproductive output even under well-watered conditions, could the CAM biochemical module be recruited into a floral regulatory network?

In this study, we build upon Holtum *et al.* (2021) and present *C. cachinalensis* as an excellent new experimental system to investigate the diversity of C₃ + CAM behavior, informed by the plant's ecology and life history. We sequenced and annotated a chromosome-level genome and performed a series of experiments designed to induce CAM under drought as well as to monitor CAM under well-watered conditions during the transition to flowering and fruiting. We reproduced the results of Holtum *et al.* (2021) and identified the core CAM gene homologues via gene expression analyses. We also demonstrated weakly increased

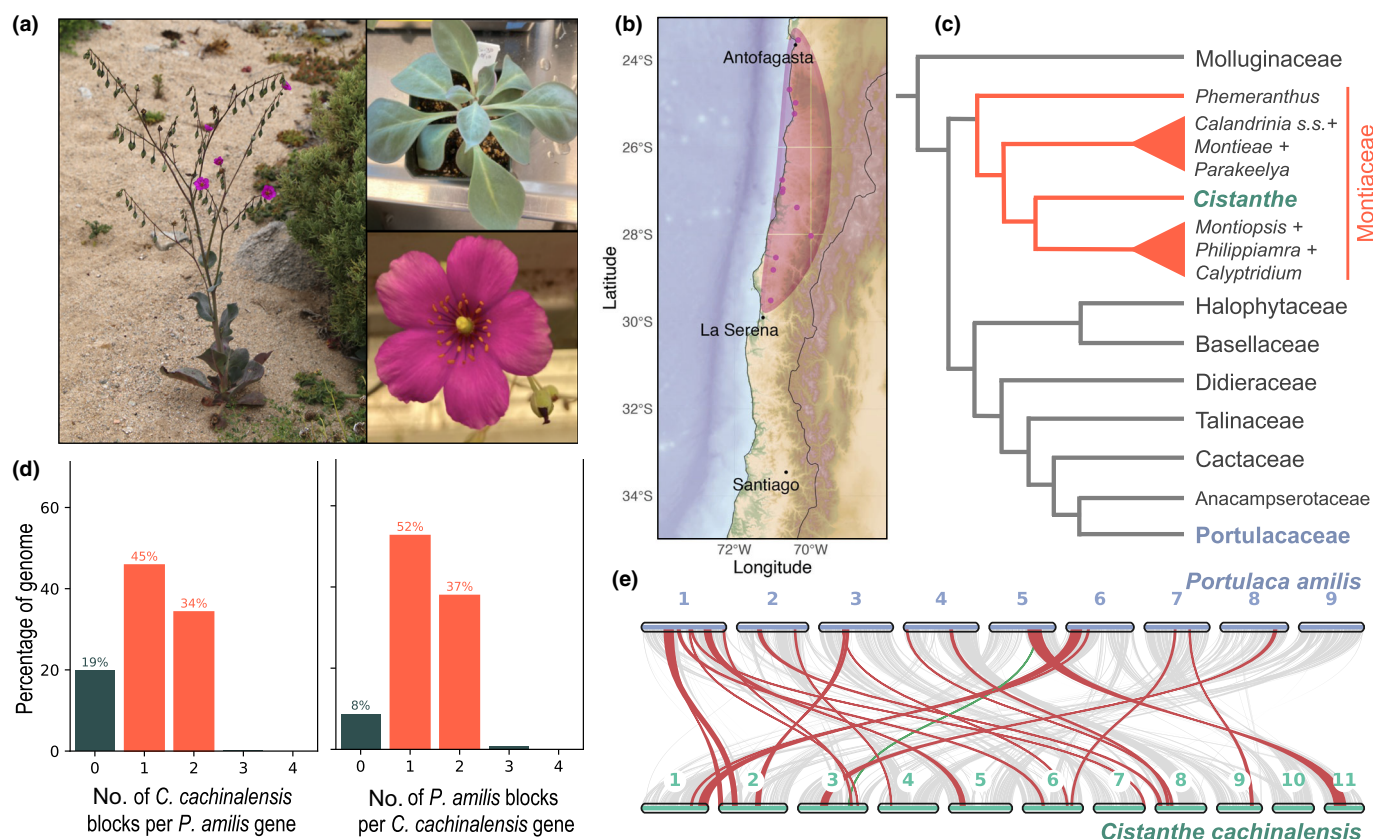


Fig. 1 Overview of *Cistanthe cachelensis* and its genome. (a) *Cistanthe cachelensis* is a desert annual C_3 + crassulacean acid metabolism (CAM) plant that blooms robustly in years with higher-than-usual precipitation, along with other mass-blooming species in its (b) native Atacama Desert region in Northern Chile. Points on the map are filtered occurrence data from the Global Biodiversity Information Facility (2022). (c) The genus *Cistanthe* is in the family Montiaceae, highlighted in the orange branches of a cladogram of the suborder Portulacineae and sister lineage Molluginaceae. (d) Syntenic depth analyses of *C. cachelensis* and *Portulaca amilis* genomes show that 34% and 37% of the *P. amilis* and *C. cachelensis* genomes had two syntenic regions, respectively. This suggests that these taxa share an ancestral whole-genome duplication. (e) The syntenic blocks between the two genomes that include core CAM genes are visualized in red, whereas the rest are visualized in gray. One gene that did not map to a similar region to the rest of its block from *P. amilis* was *PPC-1E1c* (visualized in green), the copy of the *phosphoenolpyruvate carboxylase* gene used in CAM in *C. cachelensis* and *P. amilis*.

expression of CAM in reproductive phases under well-watered conditions. Gene expression analyses partly support our physiological measurements, and motif enrichment analyses demonstrate shared transcription factor (TF) binding sites with known CAM genes and other gene clusters upregulated during flowering/fruiting, suggesting that the CAM pathway may be partly coregulated with reproduction.

Materials and Methods

Plant materials and growth conditions

Cistanthe cachelensis (Phil.) Peralta & D.I. Ford specimens were collected from the Atacama region, Chile, in 2017 and propagated at James Cook University, Australia. Seeds from the propagation were cold-stratified, and then germinated on regular potting soil at Yale University. Half were cultivated in a 12-h photoperiod chamber; the other half were cultivated in a 14-h photoperiod chamber. Plants in both chambers were exposed to 200 μmol of light during the photoperiods and grown under

ambient relative humidity and CO_2 . The temperature was maintained at 25°C during the day and 20°C at night.

Life history and gas exchange measurements

To characterize the typical lifespan and growth of *C. cachelensis*, we examined a total of 115 plants split between the 12- and 14-h chambers every 2 d to count the number of leaves, buds, flowers, and fruits. The dates of germination, reproductive stalk emergence, and flowering were recorded. In the 14-h chamber, we measured gas exchange in one pre-flowering plant under a well-watered condition and one under drought, to check that the plant was in fact showing a physiological response to our drought condition. We used the LI-6800 Portable Photosynthesis System (Li-Cor Biosciences, Lincoln, NE, USA) with a clear-top chamber, and spot-measured gas exchange in leaves across six time points (4-h intervals). The CO_2 assimilation rate ($\mu\text{mol m}^{-2} \text{s}^{-1}$) was adjusted for the total leaf area in the chamber head. We collected gas exchange measurements from four additional post-flowering individuals in well-watered conditions

to examine the CO₂ assimilation rate at the post-flowering life history stage. Due to the availability of chamber space, we conducted this experiment with a nighttime temperature of 15°C, instead of 20°C.

Genome and whole-transcriptome sequencing

Leaf tissue (5.0 g) from one juvenile individual was flash-frozen with liquid nitrogen after a 72-h dark treatment to minimize secondary metabolites, and then sent to the Genomics and Cell Characterization Core Facility (GC3F) at University of Oregon for high-molecular-weight (HMW) DNA extraction and subsequent PacBio HiFi library preparation and sequencing. Leftover material was used for Hi-C library preparation and sequencing. Furthermore, leaf, root, bud, and flower tissue were collected in four separate tubes, with two additional tubes containing all tissue types from both day- and nighttime and flash-frozen (*c.* 0.3–0.5 g per sample). Total RNA was extracted, and Iso-Seq libraries were prepared from these samples to sequence full-length mature mRNA from the whole transcriptome.

To assemble a high-quality genome, we chose PacBio HiFi and Hi-C sequencing. High-molecular-weight DNA extraction and library preparation were performed at the University of Oregon (GC3F). The genomic library was sequenced across three PacBio Sequel II SMRT cells based on prior genome-size estimates from flow cytometry ($2n = 1.35 \text{ Gbp} \pm 10\%$; this resulted in *c.* 150× depth). The Phase Genomics Hi-C library was also prepared at GC3F and sequenced on the Illumina Nova-Seq 6000 (2 × 150-bp paired-end reads).

Additionally, to obtain full-length transcripts for high-quality annotation of the genome, we sequenced the whole-transcriptome with Iso-Seq. For this, we extracted high-quality RNA from six samples of flash-frozen tissue by grinding them with a mortar and pestle in liquid nitrogen, and using a modified CTAB/TRIzol/sarkosyl-based protocol from Jordon-Thaden *et al.* (2015; 'Option 2'). This resulted in an average yield of 34 000 ng of total RNA across samples. The Yale Center for Genome Analysis (YCGA) prepared an Iso-Seq library from these extractions and sequenced the library on a PacBio Sequel II flow cell.

Genome assembly and annotation

Subreads from HiFi sequencing were collapsed into circular consensus sequence (CCS) reads before demultiplexing/quality-filtering with the PacBio CCS 6.4.0 (<https://github.com/PacificBiosciences/ccs>) pipeline. We used HiFIASM v.0.19.8 (Cheng *et al.*, 2021, 2022) to generate a *de novo* assembly combining HiFi and Hi-C reads, and YAHS (Zhou *et al.*, 2023) for further scaffolding. Hi-C reads were mapped to the draft assembly using BWA-MEM (parameters -A1 -B4 -E50 -L0; Li, 2013). The scripts 'perl filter_five_end.pl' and 'perl two_read_bam_combiner.pl' from Arima Genomics Mapping Pipeline v.03, as well as SAMTOOLS (Danecek *et al.*, 2021), were used to process and sort the Hi-C read alignments. We used JUICER (Durand *et al.*, 2016) to generate Hi-C contact maps, scripts from

<https://bitbucket.org/bredeson/artisanal/src/master/src/> to prepare Hi-C map visualization, and JUICEBOX (Robinson *et al.*, 2018) to further edit the scaffolds. Assembly completeness was assessed with BUSCO v.5.7.0 against eudicots_odb10.1 (Manni *et al.*, 2021).

Iso-Seq raw reads were preprocessed with the Pacbio Iso-Seq pipeline to generate CCS reads and filtered for full-length non-chimeric (FLNC) reads. We converted the FLNC reads into FASTQ format with SAMTOOLS (Danecek *et al.*, 2021) and mapped them with MINIMAP2 using the Iso-Seq preset (Li, 2018). The final assembly was soft-masked for repeats using RED (Girgis, 2015) and annotated using BRAKER3 (Gabriel *et al.*, 2024) with hits from ORTHODB v.10 for Viridiplantae (Kriventseva *et al.*, 2019) and the mapped Iso-Seq data. Functional annotation was performed with INTERPROSCAN v.5.67-99 (Jones *et al.*, 2014). We used EDTA (Ou *et al.*, 2019) to identify transposable elements (Tes).

Comparative genomic analyses

We compared the annotated genome of *C. cachinalensis* to the most closely related published genome, *C₄ + CAM* species *Portulaca amilis* (Gilman *et al.*, 2022). We assessed the degree of synteny between the two species at least *c.* 45 million years divergent (Ara-kaki *et al.*, 2011) using MCSCAN (Wang *et al.*, 2012) and the JCVI PYTHON library (Tang *et al.*, 2024). We used ORTHOFINDER (Emms & Kelly, 2019) to perform orthology inference and identify orthologs for all photosynthetic genes. We collected proteomes for 11 angiosperm species (*P. amilis*, Gilman *et al.*, 2022; *Ananas comosus*, Ming *et al.*, 2015; *Amaranthus hypochondriacus*, Lightfoot *et al.*, 2017; *Arabidopsis thaliana*, NCBI GenBank assembly GCA_000001735.2; *Brachypodium distachyon*, International Brachypodium Initiative, 2010; *Helianthus annuus*, Badouin *et al.*, 2017; *Kalanchoë fedtschenkoi*, Yang *et al.*, 2017; *Phalaenopsis equestris*, Cai *et al.*, 2015; *Sorghum bicolor*, Paterson *et al.*, 2009; *Vitis vinifera*, Shi *et al.*, 2023; *Zea mays*, NCBI GenBank assembly GCA_902167145.1) and clustered them with *C. cachinalensis* protein sequence using DIAMOND (ultrasensitive mode, option 'MSA'; Buchfink *et al.*, 2021). We checked that the resulting species tree was congruent with the currently accepted phylogenetic relationships and used the defined orthogroups to identify homologs in *C. cachinalensis*. We also inspected the trees manually to assess orthology between specific copies more precisely. In addition, we retrieved coding sequences from a denser sampling of CAM and non-CAM lineages for numerous available photosynthetic genes from Gilman *et al.* (2022) and added the *PEPC* gene sequence from *M. crystallinum* (Shen *et al.*, 2022). We used BLASTN (BLAST+ suite; Camacho *et al.*, 2009) to identify homologs for these genes in *C. cachinalensis* and aligned them with MAFFT v.7 (Katoh & Standley, 2013). Gene family trees were inferred using IQ-TREE v.2 (Minh *et al.*, 2020).

CAM experimental setup and collections

We set up two experiments: (1) comparing CAM expression levels in pre-flowering plants (vegetative or with reproductive

stalk but no open flowers) between drought and well-watered conditions and (2) comparing CAM expression levels between pre- and post-flowering stages (at least one flower is open, fruiting or not) under well-watered conditions. After germination, *C. cachinalensis* grows vegetatively, putting out leaves in its rosette; once a relatively consistent number of leaves accumulate, the plant will produce an indeterminate reproductive stalk (Supporting Information Fig. S1). Once flower buds develop and open, the plant will spend a significant portion of its life putting out more flowers and developing many fruits simultaneously until death. Therefore, we binned its life history between two stages, pre- and post-flowering, representing a clear difference in the plant's life history stages. The first experiment was conducted in the 12-h photoperiod chamber, and the second was conducted in the 14-h chamber. To induce a drought response, plants were withheld water for 10–14 d. There were at least four biological replicates for each of the four treatment $\times$ life history stage $\times$ photoperiod categories from the two experiments. Leaf tissue was flash-frozen from all the biological replicates from six time points every 4 h (14:00, 18:00, 22:00, 02:00, 06:00, and 10:00 h) for bulk RNA extraction. Additional leaf tissue was also collected at dusk and dawn (18:00/06:00 h) for titratable acidity assays.

Titratable acidity assays

Crassulacean acid metabolism activity was quantified by assaying titratable leaf acidity at dusk and dawn and calculating its change; since malic acid is stored in the vacuole overnight and decarboxylated during the day, the difference in titratable acidity will be measurable in plants using CAM photosynthesis. To do this, we weighed and boiled 0.3–0.4 g of fresh leaf tissue in 60 ml of 20% EtOH until the volume was reduced to 30 ml. Volume was restored to 60 ml with distilled H₂O, and the boiling and refilling process was repeated; the resulting supernatant was left to cool to room temperature. Once cooled, we titrated each sample with 0.002 M NaOH to a pH of 6.5, and then to 7.0. We recorded the initial pH as well as the total microliters of the titrant (NaOH) used. Titratable acidity (in μ equivalents of H⁺) was calculated by ml volume titrant (up to pH 7.0) $\times$ molarity (mM) of titrant and divided by initial leaf sample weight (g), to give μ mol of H⁺ per gram of fresh weight (μ mol H⁺ g⁻¹ FW). The difference between dawn and dusk leaf collection per sample, Δ H⁺, or the change in titratable acidity, is the measure for CAM activity.

RNA extractions and sequencing

We extracted total RNA from 96 leaf samples (four biological replicates for each of the six time points across the four treatments/life stages) as we did for IsoSeq. The leaf tissues were disrupted while frozen using TissueLyser III (Qiagen) with 3-mm stainless steel beads. Quantity and quality of the extractions were measured using NanoDrop spectrophotometer (Invitrogen by Thermo Fisher Scientific, Carlsbad, CA, USA) and Agilent Bioanalyzer v.2100 (Santa Clara, CA, USA). Six samples failed to meet the concentration and RNA integrity number (RIN) score

requirements ($\geq$ RIN 8) for library preparation. The RNA libraries were prepared using Poly-A selection; here, three additional samples did not meet the required concentration post-PCR. A total of 87 libraries were sequenced on the Illumina NovaSeq X PLUS (2 $\times$ 150-bp paired-end reads) at the YCGA, requesting *c.* 25 M reads per sample. All but one library was successfully sequenced, resulting in RNA-Seq data from 86 samples with each time point $\times$ treatment represented by three or four biological replicates (Table S1).

RNA-Seq read mapping and abundance quantification

For all the sequenced RNA-Seq samples, we used TRIMMOMATIC v.0.39 (Bolger *et al.*, 2014) to trim out the adapter sequences, and FASTQC (Andrews, 2010) to quality control the fragments. Then, we used STAR v.2.7.11 (Dobin *et al.*, 2013) to map the reads from each sample to our assembled genome (options '*--outFilterMismatchNmax 2*', '*--outSAMtype BAM SortedByCoordinate*'). To quantify transcript abundance per gene, we quantified the number of mapped reads to each gene using the program FEATURECOUNTS (implemented in SUBREAD v.2.0.3; Liao *et al.*, 2013, 2014) with the '*-g gene_id*' option. If a given read was mapped to multiple gene regions, a fraction (1/*x*) of the total number of aligned regions (*x*) were assigned per read per gene (option '*-M --fraction*'). Quality control (QC), mapping, and count statistics were aggregated with MULTIQC (Ewels *et al.*, 2016). To quantify expression levels for each gene, we calculated transcripts per million (TPM), which normalizes for sequencing depth per sample.

Differential gene expression analysis

The count dataset from FEATURECOUNTS was imported into R v.4.4.0 (R Core Team, 2024), and normalized using the median of ratios method in the DESeq2 (Love *et al.*, 2014) package. This, along with mapping statistics from STAR, revealed one outlier sample, which was removed from further analysis. We used principal component analysis and hierarchical clustering methods to quality control at the gene and sample level and explore variation in the data. To test for significant differential gene expression between the treatments and life stages in the two experiments across time ('Drought vs Well-Watered' and 'Post-flowering vs Pre-flowering'), we used a likelihood ratio test with the full formula '*design = ~ Time + {Treatment, Stage}*' with the reduced formula '*reduced = ~ Time*'. We also tested significant diurnal gene expression (i.e. the effect of time) for each category (Drought, Watered, Post-Flowering, and Pre-Flowering) separately, with the formula '*design = ~ Time*' with the reduced formula '*reduced = ~ 1*'. We used ASHR (Stephens *et al.*, 2023) implemented in DESeq2 to shrink the log₂fold change to control for its inflation in smaller values. Our significance threshold was an adjusted (Benjamini–Hochberg) *P*-value < 0.01 , and an absolute value of log₂foldchange > 0.58 . We used annotated Gene Ontology (GO) IDs to retrieve GO terms related to biological processes using the ANNOTATIONDBI package (Pagès *et al.*, 2024) to perform GO enrichment analysis with the

enrichr function and gene set enrichment analysis (GSEA) with the GSEA function from the CLUSTERPROFILER package v.4.12.0 (Wu *et al.*, 2021).

Expression profile clustering and motif analysis

To identify similar expression profiles of differentially expressed (DE) genes between treatment/life stages and across time, we used the degPatterns function in the R package DEGREPORT (Pantano, 2024) to cluster genes via similarity. For diurnally DE genes that clustered together with core CAM genes, we extracted 2-kb up- and downstream to capture regulatory regions (with custom scripts using BEDTOOLS; Quinlan & Hall, 2010). We then used XSTREME (Grant & Bailey, 2021) in the MEME suite (Bailey *et al.*, 2015) v.5.5.7 with the default settings, which utilizes several of its own functions to discover (MEME and STREME; Bailey & Elkan, 1994; Bailey, 2021, respectively), annotate (TOMTOM; Tanaka *et al.*, 2011), and identify enriched motifs (SEA; Bailey & Grant, 2021) within clusters, generating control sequences from regulatory regions of 1000 randomly sampled genes. We used the 2022 JASPER CORE DNA nonredundant plant motif database (Castro-Mondragon *et al.*, 2021) as the index of known motifs. XSTREME also clusters motifs via similarity (which we will refer to as a ‘family’ of clustered motifs) as its final output. We also ran AME (McLeay & Bailey, 2010) for additional motif enrichment analysis.

Results

Genome assembly, annotation, and synteny

The assembled haploid *C. cachtinalensis* genome is 647.33 Mbp in size (which is consistent with our estimate of $2n = 1.35 \text{ Gbp} \pm 10\%$), composed of *c.* 46% repeats (Table S2), and highly contiguous (L50 = 6 scaffolds; N50 = 44 Mb; 90% of the genome exists in 940 contigs). A total of 28 298 unique protein-coding genes were annotated, which represents a completeness of 97.0% based on BUSCO v.5.7.0 eudicots_odb10.1 database (C: 97.0% (S: 80.4%, D: 16.6%), F: 0.2%, M: 2.8%, *n*: 2326). The 11 largest scaffolds represent $x = 11$ chromosomes ($2n = 2x = 22$ chromosomes; Fig. S2), which is consistent with the signal intensity pattern from the Hi-C contact map (Fig. S3) as well as previous surveys of base chromosome numbers in Montiaceae (Marinho *et al.*, 2019). Core CAM genes are scattered across these 11 chromosomes (Fig. S4). Specifically for the *PEPC* gene, we found the *PPC-1E1a*, *PPC-1E1c*, *PPC-1E2* (Fig. S5), and *PPC-2* copies, which are consistent with copies found previously in other Montiaceae species (Christin *et al.*, 2014).

Comparisons between the *C. cachtinalensis* genome and the *P. amilis* genome showed that there are large syntenic blocks that are preserved relative to the evolutionary distance and chromosome number variation, which includes many genes that are in the CAM photosynthetic pathway (Fig. 1e). Furthermore, analyzing the syntenic depth between the two genomes revealed that 37% of the genes in the *C. cachtinalensis* genome have two

homologs in the *Portulaca* genome; inversely, 34% of *P. amilis* genes have two homologs in the *Cistanthe* genome (Fig. 1d). This reciprocal pattern suggests a whole-genome duplication in a common ancestor, with both genomes losing the duplicated genes at roughly the same rate. This is consistent with a previous study that inferred a whole-genome duplication at the base of the sub-order Portulacineae (Yang *et al.*, 2018).

Induction of CAM phenotypes

Our drought treatment produced a physiological effect on the plants. Twenty-four-hour monitoring of gas exchange showed that individuals under drought have a positive carbon assimilation rate at night (in 14-h day length; Fig. 2a). In general, we found that *C. cachtinalensis* exhibited very low rates of gas exchange in our experiments, whether well-watered or droughted (Fig. 2a). Pre-flowering plants that were subjected to drought significantly increased their nighttime acid accumulation (the CAM phenotype) compared with well-watered plants, as measured by the change in titratable acidity between dawn and dusk (Fig. 2b; one-sided Wilcoxon signed-rank sum exact test, P -value = $5.744e^{-5}$). Moreover, well-watered, post-flowering plants also exhibited, on average, a positive nighttime carbon assimilation rate (Fig. 2a), and they increased their nighttime acid accumulation compared with pre-flowering plants (Fig. 2b; P -value = 0.0603), although the difference in mean ΔH^+ was not as large as the drought vs well-watered comparison from the previous experiment.

Transcriptome assembly and gene abundance

Across 87 RNA libraries, an average of 33.23 million reads per sample were sequenced. We recovered an average of 24.53 million uniquely mapped reads (73.87%) with STAR; 48.21 million aligned sequences (62.72%) were assigned a feature with FEATURE-COUNTS (this includes multimapped reads from the STAR output; Dataset S1). A total of 25 504 genes (out of 28 298 annotated genes) were present in the final gene-count dataset. Of these genes, 493 genes were annotated as orthologous copies of genes associated with photosynthesis (Gilman *et al.*, 2022; Dataset S2).

In both experiments, the core genes in the CAM pathway (Fig. 3a) displayed consistent results: *C. cachtinalensis* individuals under the drought treatment showed increased abundance and CAM-like expression patterns compared with those under the well-watered treatment, and post-flowering plants showed similar patterns when compared to pre-flowering ones with slightly less overall expression levels with some exceptions (Fig. 3b,c). Despite having CAM-like expression patterns, and sometimes significant diurnal expression in one stage and not the other (see ‘Differential expression patterns’ in the Results section), we saw no significant effect of life history stage to the expression level differences of these core CAM genes, even when averaging across time points. Importantly, we found that the *1E1c* copy of the *PEPC* gene was utilized in the CAM pathway in the drought/well-watered comparison; increased expression of *PPC-1E1c* in early evening in the drought condition is a signature CAM phenotype

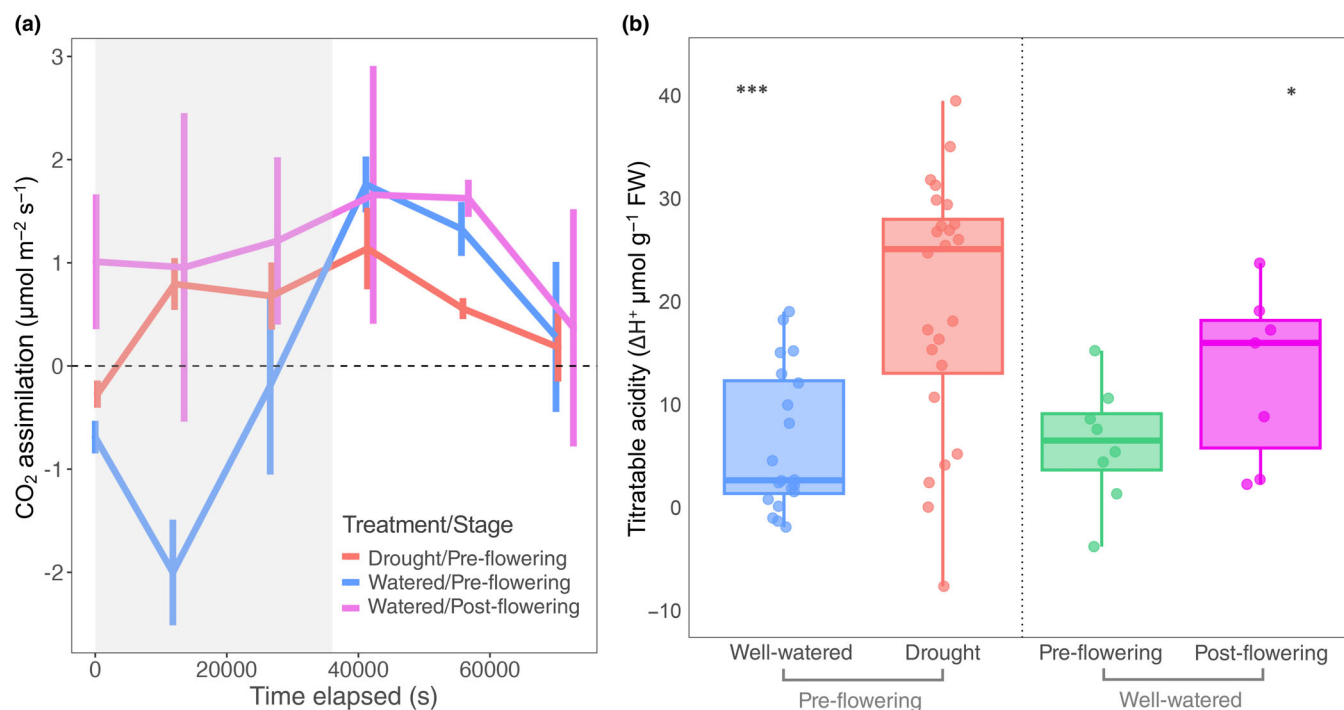


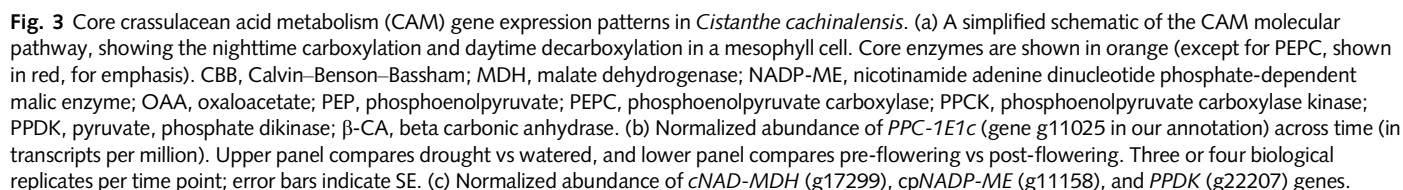
Fig. 2 Crassulacean acid metabolism (CAM) physiology in *Cistanthe cachinalensis*. (a) CO_2 assimilation rate ($\mu\text{mol m}^{-2} \text{s}^{-1}$) measured across six timepoints over 24 h (time elapsed shown in seconds) in an individual under drought at the pre-flowering (vegetative) stage (red), a separate individual under a well-watered treatment at the pre-flowering stage (blue), and four additional individuals under a well-watered treatment at the post-flowering stage (magenta). Error bars are $\pm\text{SD}$ taken directly from the LI-6800 machine, which was used to measure the assimilation rate for the drought/pre-flowering and watered/pre-flowering lines ($n = 1$ for both lines), and $\pm\text{SD}$ calculated from the four assimilation rate measurements taken for the watered/post-flowering line ($n = 4$). Time period with lights off (10 h) is shaded in gray. (b) Titratable acidity (in μ equivalents of H^+ ; $\Delta\text{H}^+ \mu\text{mol g}^{-1} \text{FW}$) assayed among leaf samples of *C. cachinalensis* in (1) well-watered treatment and drought treatment (pre-flowering individuals), as well as in (2) pre- and post-flowering stages (grown under well-watered conditions). This shows nocturnal acid accumulation (the difference between dawn and dusk collection per sample), which is a measure of CAM activity. Boxplots show median and interquartile range (whiskers = $1.5 \times$ interquartile range). Asterisks indicate significance from Wilcoxon signed-rank sum exact test (*, $P < 0.1$; ***, $P < 0.001$).

(Fig. 3b). The post-flowering plants also had CAM-like expression profiles in *PPC-1E1c*, although the highest peak was present right before nighttime rather than early evening; two samples in pre-flowering had spikes of *PPC-1E1c* expression in the night, each at different time points (16 and 20 Zeitgeber hours; Fig. 3b), contributing to the high variation and large error bars. We identified that the cytosomal *Malate dehydrogenase* gene copy (*cNAD-MDH1-1*) and the *malic enzyme* gene copy localized in the chloroplast (*cpNADP-ME4-1*) are utilized in the CAM pathway (carboxylation and decarboxylation, respectively; Fig. 3c). *Cistanthe cachinalensis* using nicotinamide adenine dinucleotide phosphate-dependent malic enzyme (NADP-ME) rather than NAD-ME (Fig. S6) for decarboxylation is consistent with the CAM biochemistry seen in *Portulaca* (Gilman *et al.*, 2022; Moreno-Villena *et al.*, 2022). Notably, both *phosphoenolpyruvate carboxylase kinase* (*PPCK*) and *phosphoenolpyruvate carboxykinase* (*PCK*) genes were hardly expressed in any plant between the two experiments (Fig. S6). This suggests that these enzymes are not necessary in the CAM pathway and that PEPC does not need phosphorylation in *C. cachinalensis*. This may be due to a mutation in the *C. cachinalensis* *PPC-1E1c* at the Position 495 arginine to glutamine (relative to *Portulaca*; Fig. S7), a position

where a mutation from arginine to aspartic acid in the *K. fedtschenkoi* *PEPC1* gene has shown to significantly increase PEPC1 activity without phosphorylation (Yang *et al.*, 2017). Lastly, the expression of *pyruvate, phosphate dikinase* (*PPDK*) also followed the overall pattern: increased abundance and CAM-like expression patterns in drought-treated and post-flowering plants compared with well-watered and pre-flowering ones, but with less overall expression levels in post-flowering compared with drought. Normalized abundance plots from all other core genes and gene copies are available in Fig. S6.

Differential expression patterns

Many genes showed significant differential expression when tested for the effect of treatment or life stage across time in both experiments (4769 genes in 'Drought vs Watered', of which 144 were photosynthetic pathway genes; 1256 genes in 'Post-Flowering vs Pre-Flowering', of which 26 were photosynthetic pathway genes; Figs S8–S11; Dataset S3). *PPC-1E1c* ($P_{\text{adj}} = 7.47\text{e}^{-5}$), *cpNADP-ME4-1* ($P_{\text{adj}} = 1.99\text{e}^{-4}$), and *cNAD-MDH1-1* ($P_{\text{adj}} = 4.01\text{e}^{-4}$) were differentially expressed (DE) in the 'Drought vs Watered' experiment, while other photosynthetic



pathway genes were significantly DE in the 'Post-Flowering vs Pre-Flowering' experiment (Fig. S12). Around half of the DE genes in the 'Post-Flowering vs Pre-Flowering' comparison were shared with those in the 'Drought vs Watered' comparison (568

genes). Among these 568 shared DE genes, 512 were up- or downregulated in the same direction in both experiments, and only 56 were up- and downregulated in the opposite direction (Dataset S4). Furthermore, among the 568 genes, we found TFs

Table 1 Select examples from 568 overlapped differentially expressed genes between the 'Drought vs Watered' and 'Post-Flowering vs Pre-Flowering' comparisons in *Cistanthe cachinalensis*¹.

Gene	Gene IDs	Primary role	Citation	Regulation ²
AT-hook motif nuclear-localized protein	g25175	DNA-binding protein, involved in regulation of growth and development	Zhao <i>et al.</i> (2014)	Up
Auxin response factor (ARF)	g16661	Regulates auxin-mediated transcription, crucial for plant growth and development	Liscum & Reed (2002); Van Ha <i>et al.</i> (2013)	Down
B3 domain-containing protein, REM16-like	g3466	Involved in development, expressed during early stages of flowering	Mantegazza <i>et al.</i> (2014)	Up
Basic helix–loop–helix (bHLH) leucine zipper transcription factor	g14561	Regulates various developmental and metabolic processes	Heim <i>et al.</i> (2003)	Down
DOF zinc finger protein	g620	Transcription factor involved in vascular development and photoperiodic flowering	Konishi & Yanagisawa (2007); Fornara <i>et al.</i> (2009)	Down
Ethylene-responsive transcription factors (ERF/AP2)	g9235, g19800, g21117	Binds to ethylene-responsive elements, involved in drought and other stress response	Fujimoto <i>et al.</i> (2000); Sakuma <i>et al.</i> (2002)	Down
Homeobox-leucine zipper proteins	g4536, g21444, g25444	Regulates developmental processes, involved in cold stress tolerance, abscisic acid signaling	Henriksson <i>et al.</i> (2005); Cabello <i>et al.</i> (2012); Valdés <i>et al.</i> (2012)	Up ³
MADS-box transcription factor	g19958	Regulates flower development and reproductive processes	Rounsley <i>et al.</i> (1995)	Up
MYB family transcription factors, HRS/PHR-like	g6103, g14400	Transcription factors involved in abscisic acid signaling and phosphate starvation response	Wu <i>et al.</i> (2012); Yan <i>et al.</i> (2014); Nagatoshi <i>et al.</i> (2023)	Up (g6103)/ Down (g14400)
Squamosa promoter-binding-like proteins	g2304, g12774, g21618	Transcription factors regulating flowering and reproductive processes	Klein <i>et al.</i> (1996)	Up (g21618)/ Down (g2304, g12774)
TIFY/JAZ family protein	g6762	Jasmonate signaling repressors, involved in biotic and abiotic stress responses	Chini <i>et al.</i> (2007, 2017)	Up
Absciscic acid-/WDS-induced protein	g3570	Induced by abscisic acid and drought stress, involved in water-deficit stress tolerance	Padmanabhan <i>et al.</i> (1997)	Up
Aquaporin transporters	g13245, g1437	Water channel proteins, regulate water transport across cell membranes	Maurel & Chrispeels (2001); Afzal <i>et al.</i> (2016)	Down
Cysteine-rich and transmembrane domain-containing protein	g14486	Involved in stress response/signaling and plant development	Lieber <i>et al.</i> (2011); Xu <i>et al.</i> (2018)	Up
DCD domain-containing protein, NRP-like	g16662	Involved in stress signaling that mediate cell death, associated with abscisic acid signaling	Costa <i>et al.</i> (2008); Zhu <i>et al.</i> (2018)	Up
DEAD-BOX-dependent RNA helicase 50	g16819	Involved in RNA metabolism, splicing, and translation under stress	Li <i>et al.</i> (2023)	Up
Late embryogenesis abundant (LEA) proteins	g24748, g11895, g9133	Seed embryogenesis and drought-stress response and tolerance	Tunnacliffe & Wise (2007)	Up (g9133), Down (g24748, g11895)
Peroxidases	g5240, g5324, g5325, g10678, g17827	Involved oxidative stress response and stress tolerance	Mittler (2002)	Down
Proline-rich extensin	g16111	Cell wall structural protein necessary for growth and development	Saha <i>et al.</i> (2013)	Down
Small heat shock proteins	g5672, g17836	Molecular chaperones, protect proteins from aggregation during heat stress	Lindquist & Craig (1988)	Down

Bold gene = transcription factor. ATP, adenosine triphosphate; DCD, development and cell death; DOF, DNA binding with one finger; JAZ, Jasmonate ZIM-domain; NRP, asparagine-rich protein; PHR, phosphate starvation response; WDS, water deficit stress.

¹Genes related to stress response, development, and reproduction are highlighted here, but the full 568 gene list is in Supporting Information Dataset S4.

²'Up' or 'Down' refers to genes being upregulated or downregulated in the Drought treatment and the Post-Flowering stage when compared to the Watered treatment and the Pre-Flowering stage, respectively.

³Except for g4536 in the Drought vs Watered comparison.

and genes that are involved in general stress response, development, or reproduction; we highlight these genes in Table 1.

Of the 568 overlapped DE genes between the two experiments, 11 were photosynthetic pathway genes (Fig. 5). Six genes related to starch metabolism were all upregulated in the drought treatment and post-flowering stage, as well as a gene in the photorespiration pathway. Conversely, a gene belonging to proton transport and another in the circadian/light response pathway were downregulated in both the drought treatment and the post-flowering stage (compared with the watered treatment and pre-flowering stage, respectively). Two genes were up- and downregulated in opposite directions: *RuBisCO small subunit* gene (Calvin cycle) and *Scarecrow-like* gene (circadian/light response).

Many genes also showed significant differential expression with respect to time (i.e. exhibit diurnal patterns; $P_{\text{adj}} < 0.01$) in each treatment and life stage (5028 genes in the drought treatment, 4688 in watered, of which 2352 overlap; 2356 in the post-flowering stage, 2321 in pre-flowering, of which 1146 overlap; Figs 4a, S10). *PPC-IE1c* was DE across time in drought conditions ($P_{\text{adj}} = 0.0027$) and in the post-flowering stage ($P_{\text{adj}} = 7.49\text{e}^{-4}$). Additionally, *cpNADP-ME4-1* was also diurnally DE in both the drought treatment ($P_{\text{adj}} = 9.29\text{e}^{-7}$) and post-flowering stage ($P_{\text{adj}} = 2.11\text{e}^{-11}$), while *cNAD-MDH1-1* was diurnally DE only in the post-flowering stage ($P_{\text{adj}} = 2.87\text{e}^{-6}$). Various other photosynthetic pathway genes were significantly DE in both the drought treatment and the post-flowering stage, but not in the watered treatment or pre-flowering stage; a number of these genes were related to starch metabolism and circadian/light response (Fig. S13).

Moreover, we clustered genes via expression profile similarity to look at co-expression of all diurnally DE genes (Fig. 4a; Dataset S5). In both the drought treatment and post-flowering stage, we found that many photosynthetic pathway genes were co-expressed with each other (in clusters with other genes), although the co-expression could not be explained by pathway identity (Fig. 4a). We also visualized the expression profile clusters for all DE genes across the two experiments in Fig. S14, and just the DE photosynthetic pathway genes that belong to the 'Drought', 'Post-Flowering', and 'Drought vs Watered' sets (141 genes, 118 genes, and 144 genes, respectively), which can be found in Figs S15 and S16. Additionally, results from GO enrichment and GSEA can be found in Figs S17 and S18, with terms related to photosynthesis, growth, or stress response found across DE gene sets (Fig. S18).

Motif enrichment

We looked at regulatory regions of all diurnally DE genes that were co-expressed with *PPC-IE1c* in the post-flowering stage (Cluster 2; Fig. 4a) and the drought treatment (Cluster 10; Fig. 4a) to discover motifs and test for enrichment. Overall, we found (1) the same circadian rhythm associated motifs enriched in the post-flowering cluster that were enriched in a previous study with genes co-expressed with *PPC-IE1c* in *Portulaca* under drought (Gilman *et al.*, 2022). (2) We found three enriched

regulatory motifs shared between the post-flowering and drought clusters that are associated with stress-related gene regulation, plant development, and photoperiodic flower regulation. (3) Between the post-flowering and drought clusters, we found three motifs that are enriched in the post-flowering but not in the drought.

Five motif families (from 12 discovered motifs and nine motif matches from the JASPAR CORE dataset) were enriched in the post-flowering cluster (Fig. 4b; Table 2). Outputs from XSTREME and AME were used to confirm that all five motifs are in the regulatory regions of *PPC-IE1c*. One of the enriched discovered motifs (1-AAAAATATCTT; Table 2) belongs to the MYB-related evening element (EE) family, which has been associated with TFs involved in circadian regulation (e.g. RVE8, LHY, and CCA; Carré & Kim, 2002) and has been implicated in CAM regulation (Gilman *et al.*, 2022). Another enriched motif (MEME-2; Table 2) belongs to the ERF/DREB family predicted to be involved in stress-related gene expression (Fujimoto *et al.*, 2000; Sakuma *et al.*, 2002). XSTREME clustered this motif with the also enriched GAGA-repeats (MEME-1, -9, and -5; Table 2; Dataset S6), which are binding sites for plant-specific barley B recombinant/basic pentacysteine (BBR/BPC) TFs involved in plant development (Sahu *et al.*, 2023). Finally, the DOF zinc finger protein-related motifs (MEME-6; Table 2) were enriched, which have been shown to be associated with photoperiodic flowering response regulation in *Arabidopsis* (Fornara *et al.*, 2009).

In the drought set, we found five enriched motif families (from eight discovered motifs and one motif match from the JASPAR CORE dataset), all of which are found in the regulatory regions of the *PPC-IE1c* gene, and three of which were also enriched in the post-flowering cluster (Fig. 4b; Table 2). The enriched MYB-related TRP5 motif is associated with TFs primarily involved in telomeric DNA-binding (Hwang *et al.*, 2005). Another enriched motif (2-AGAACTAT; Table 2) is associated with NLT8 (an NAC TF) involved in repressing flowering and germination in *Arabidopsis* (Kim *et al.*, 2007, 2008).

We also looked at regulatory regions of all of the diurnally DE genes that were co-expressed with the daytime-expressed CAM pathway genes *cpNADP-ME4-1* in post-flowering (Cluster 5) and drought (Cluster 1) and *cNAD-MDH1-1* in the post-flowering cluster (Cluster 8; Fig. 4a). In all these clusters, we found the same three motifs enriched (and in the regulatory regions of the genes) as in the *PPC-IE1c* co-expressed clusters from drought and post-flowering, related to gene regulation of stress response (ERF/DREB family), development (GAGA-repeats), and photoperiodic flowering (DOF zinc finger protein-related motifs; Fig. 4b; Dataset S6). Notably, however, there were unique motifs that are only present in the post-flowering clusters but not in the drought. In the post-flowering *cpNADP-ME4-1* cluster, a novel motif was enriched (consensus sequence 'TCGGDTTCGGRTHCGG'; Dataset S6), which was in the regulatory regions of *cpNADP-ME4-1*. In the *cNAD-MDH1-1* cluster, MYB-related EE motifs (circadian) were enriched (Dataset S6), which were also enriched in the *PPC-IE1c* post-flowering cluster.

Discussion

Cistanthe cachinalensis as an additional 'model' C₃ + CAM species

Crassulacean acid metabolism photosynthesis has evolved in many dozens of lineages independently (Gilman *et al.*, 2023) and presents a diverse set of phenotypes that we still know so little about, in terms of their regulation, their potential evolutionary relationships to one another, and even their adaptive significance (Winter *et al.*, 2015; Winter, 2019; Edwards, 2023). In the laboratory, the CAM species best studied at a molecular level are *M. crystallinum* (Bohnert *et al.*, 1988), *Kalanchoë* (Hartwell *et al.*, 2016), and *Yucca* (Heyduk *et al.*, 2019), representing only a handful of CAM origins. The recent accumulation of annotated genomes and gene expression studies is opening new opportunities to study the molecular underpinnings of convergent CAM phenotypes in a more diverse set of taxa (Cai *et al.*, 2015; Ming *et al.*, 2015; Brillhaus *et al.*, 2016; Wickell *et al.*, 2021; Gilman *et al.*, 2022; Heyduk *et al.*, 2023).

Cistanthe cachinalensis is a new addition to the growing list of C₃ + CAM species available for molecular investigation and laboratory experimentation. *Cistanthe cachinalensis* will readily germinate after sufficient vernalization, is self-fertile, and sets fruit consistently, making it easily tractable to generate multiple generations in the laboratory. The species is a member of the Atacama Desert *desierto florido*, a subject of great interest across disciplines (e.g. Chávez *et al.*, 2019; Araya *et al.*, 2020). Importantly, *C. cachinalensis* belongs to an emerging 'model clade' (*sensu* Donoghue & Edwards, 2019), the Montiaceae, with increasingly detailed phylogenetic resolution and analyses of organismal evolution that will help contextualize what we learn about its CAM behavior and regulation (Ogburn & Edwards, 2015; Hancock *et al.*, 2018, 2019; A. Chomentowska, S. Dauerman, N. Heaphy, A. Carranza, L. Gaspar, I. E. Peralta, P.W. Sweeney, A. M-M, R.M. Ogburn, C.M. Guiliams, L.P. Hancock, M. Arakaki, J.A. M. Holtum & E.J. Edwards, unpublished).

Moreover, we now have a high-quality genome of *C. cachinalensis*, which is critical for further development of this species into an actual genetic model system (e.g. establishing transformation). The genome is already key for gene expression studies and comparative genomics – for example, the syntenic depth analyses in this study confirmed a whole-genome duplication in a common ancestor of *Portulaca* and *Cistanthe*, likely at the base of Portulacineae as hypothesized by Yang *et al.* (2018). Against a backdrop of significant genomic rearrangement, *C. cachinalensis* photosynthetic genes largely belong in syntenic blocks shared with *P. amilis* (Fig. 1e), a relative whose shared common ancestor likely had already established a facultative CAM cycle (Goolsby *et al.*, 2018; Gilman *et al.*, 2023). A notable exception, of course, was the *1E1c* copy of the *PEPC* gene, which did not stay in its syntenic block with other genes. Overall, the *C. cachinalensis* genome should be a valuable contribution to broader studies of comparative CAM genomics.

A diversity of CAM behavior in *C. cachinalensis*

Cistanthe cachinalensis is endemic to the Atacama Desert and adjacent areas of Chile, one of the most arid landscapes on Earth. Even though they germinate in response to rainfall, these plants may experience drought for much of their development, with effects accumulating over their short lifetime. Because they are facultative CAM plants, if they are experiencing drought around the time they flower and reproduce, they are likely also performing CAM photosynthesis. While CAM is typically considered an adaptation to increase plant water-use efficiency, this physiological module may be beneficial for other biological processes as well (Herrera, 2009). Reproduction, particularly flower and seed development, represents a substantial carbon investment (Bazzaz *et al.*, 1987), and it is possible that induction of a CAM cycle during reproduction could increase CO₂ fixation, supplementing daytime C₃ CO₂ fixation with recapturing respiratory CO₂ or even fixing small amounts of atmospheric CO₂ at night. Provided a plant is not light-limited and can subsidize the energetic costs of CAM, why would an annual plant with a functional CAM

Fig. 4 Results from differential gene expression analyses, expression profile clustering, and motif enrichment in *Cistanthe cachinalensis*. (a) Clustering of all genes with significant diurnal expression via expression profile similarity in the drought treatment (left) and post-flowering stage (right). Boxplots show median and interquartile range (whiskers = 1.5 × interquartile range) of gene expression in z-score normalized gene abundance; each individual expression profile is shown by thin lines, whereas the overall trend is represented with a thick line. Venn diagrams of genes with significant diurnal expression (determined by testing the effect of time) separately in the drought and watered treatments and their overlaps, as well as the post-flowering and pre-flowering stages and their overlaps. The numbers in parentheses represent the number of photosynthetic genes in the differentially expressed (DE) sets of genes, and the genes inside the Venn diagrams are core crassulacean acid metabolism (CAM) gene copies that were DE. These include the following: *beta carbonic anhydrase* (*BCA1-1*, *BCA1-5*); *NAD-dependent malate dehydrogenase*, *cytoplasmic* (*cNAD-MDH1-1*); *NADP-dependent malic enzyme*, *chloroplastic* (*cpNADP-ME4-1*); *phosphoenolpyruvate carboxylase* (*PPC-1E1c*); and *ribulose bisphosphate carboxylase* (*RuBisCO*) *activase* (*RCA-1*, *RCA-2*). Bar plots show the number of photosynthetic pathway genes from each corresponding cluster (left panel – Drought; right panel – Post-Flowering). The colors correspond to the photosynthetic pathway identity (as defined by Gilman *et al.*, 2022) of each gene, shown in the legend on the right. Individual cluster plots highlighted with a black box and the numbers circled on the x-axis of the bar plots correspond to clusters where core CAM genes belong. Times have been presented in AM/PM format on the x-axis due to space constraints. (b) Enriched motifs in the regulatory regions of all genes (not just the photosynthetic pathway genes) in the clusters where core CAM genes were found; *cpNADP-ME-4-1* (Cluster 1) and *PPC-1E1c* (Cluster 10) in the Drought treatment, and *PPC-1E1c* (Cluster 2), *cpNADP-ME-4-1* (Cluster 5), and *cNAD-MDH1-1* (Cluster 10) in the Post-Flowering stage. Motifs related to DNA binding with one finger (DOF) zinc finger proteins ('AAAAAAAAAAAA'), ethylene response factor/dehydration-responsive element-binding (ERF/DREB) family proteins ('CCCCCCCCCCC'), and GAGA-repeats (BBR/BPC class; 'GAGGAGAGAGA') were found to be enriched in all five clusters. In all clusters except for Drought Cluster 1 (*cpNADP-ME4-1*), there were other unique enriched motifs. The only other motif shared is a circadian MYB-related evening element motif ('AAAAATATCTCT') in Cluster 2 (*PPC-1E1c*) and Cluster 10 (*cNAD-MDH1-1*) in the Post-Flowering stage.



cycle not engage it to supplement its carbon uptake during seed production, irrespective of its water status?

Our experiments demonstrated that flowering and fruiting plants show elevated nocturnal accumulation of titratable acidity

and nocturnal CO_2 uptake relative to vegetative plants even when under well-watered conditions, although the level of induction was not as significant as their drought response (Fig. 2b). This pattern is also demonstrated in the expression of known CAM

Table 2 Enriched regulatory motifs from 2-kb up- and downstream of co-expressed genes with *phosphoenolpyruvate carboxylase (PPC-1E1c)* from (1) the post-flowering stage and (2) drought treatment experiments in *Cistanthe cachtalensis*¹.

	Consensus sequence	Motif source	Other motifs in family	Similar known motifs ²	E-value ³	Motif shared?
(1) Post-flowering	AAAAATATCTT	1-AAAAATATCTT (STREME)	RVE6 (MA1183.1), RVE5 (MA1190.1), RVE8 (MA1182.1), RVE1 (MA1184.1), RVE4 (MA1187.1), LHY (MA1185.1), CCA1 (MA0972.1), RVE7 (MA1401.1), RVE7L (MA1191.1)	RVE7 (MA1401.1), RVE4 (MA1187.1), RVE6 (MA1183.1)	3.52e ⁻²	Shared with the post-flowering cNAD-MDH1-1 co-expressed cluster
	GCAGGAGA	2-GCAGGAGA (STREME)	MEME-4 (MEME), 3-GCAGTGGACAA (STREME)	ASIL2 (MA2039.1), AT5G05550 (MA2040.1)	6.01e ⁻²	na
	GGGGGGGGGRGGRGG	MEME-2 (MEME)	MEME-1 (MEME), MEME-9 (MEME), MEME-5 (MEME), MEME-8 (MEME)	ERF057 (MA1753.1), ERF5 (MA1225.1), DREB2D (MA1250.1)	6.20e ⁻⁴⁸	Shared with all other co-expressed clusters investigated
	TTTTTTTHTTTTTT	MEME-6 (MEME)	MEME-3 (MEME)	CDF5 (MA1268.1), DOF3.6 (MA1274.1), DOF5.8 (MA1267.1)	2.20e ⁻¹¹	Shared with all other co-expressed clusters investigated
(2) Drought	TGTAAGRAGYGTGTA NHWAAACCYH	MEME-7 (MEME) TRP5 (MA1802.1) (SEA)	na 3-AAACCCTAAACCCC (STREME)	na TRP2 (MA1356.1), AT4G12670 (MA1354.1), TRP1 (MA1352.1)	3.30e ⁻⁹ 3.43e ⁻³	na na
	AGAAACTAT AAAAAAAAABAAAAA	2-AGAAACTAT (STREME) MEME-5 (MEME)	1-ACACCTCTYWCAT (STREME) MEME-3 (MEME)	NTL8 (MA1678.2), CDF5 (MA1268.1), DOF3.6 (MA1274.1), SGR5 (MA1159.1)	5.37e ⁻² 3.20e ⁻¹⁶	na Shared with all other co-expressed clusters investigated
	CCCCCCCCCCCCCCCC	MEME-2 (MEME)	MEME-4 (MEME)	DREB2D (MA1250.1), ERF10 (MA1240.1), ERF104 (MA1239.1)	1.40e ⁻⁷²	Shared with all other co-expressed clusters investigated
	GAGAGAGAGAGAGAG	MEME-1 (MEME)	na	BPC5 (MA1403.1), BPC6 (MA1402.1), BPC1 (MA1404.1)	1.20e ⁻⁹²	Shared with all other co-expressed clusters investigated

na, not applicable.

¹A full result table, including all examined clusters, is available in Supporting Information Dataset S6, and the entire XSTREME output is uploaded on GitHub (<https://github.com/anrilam/Cistanthe-RNASeq>).

²Similar known motifs are results from TOMTOM, but only those with similarity E-values of < 1.0 to the discovered motif are shown.

³E-values are from the motif discovery step from MEME, STREME, or SEA.

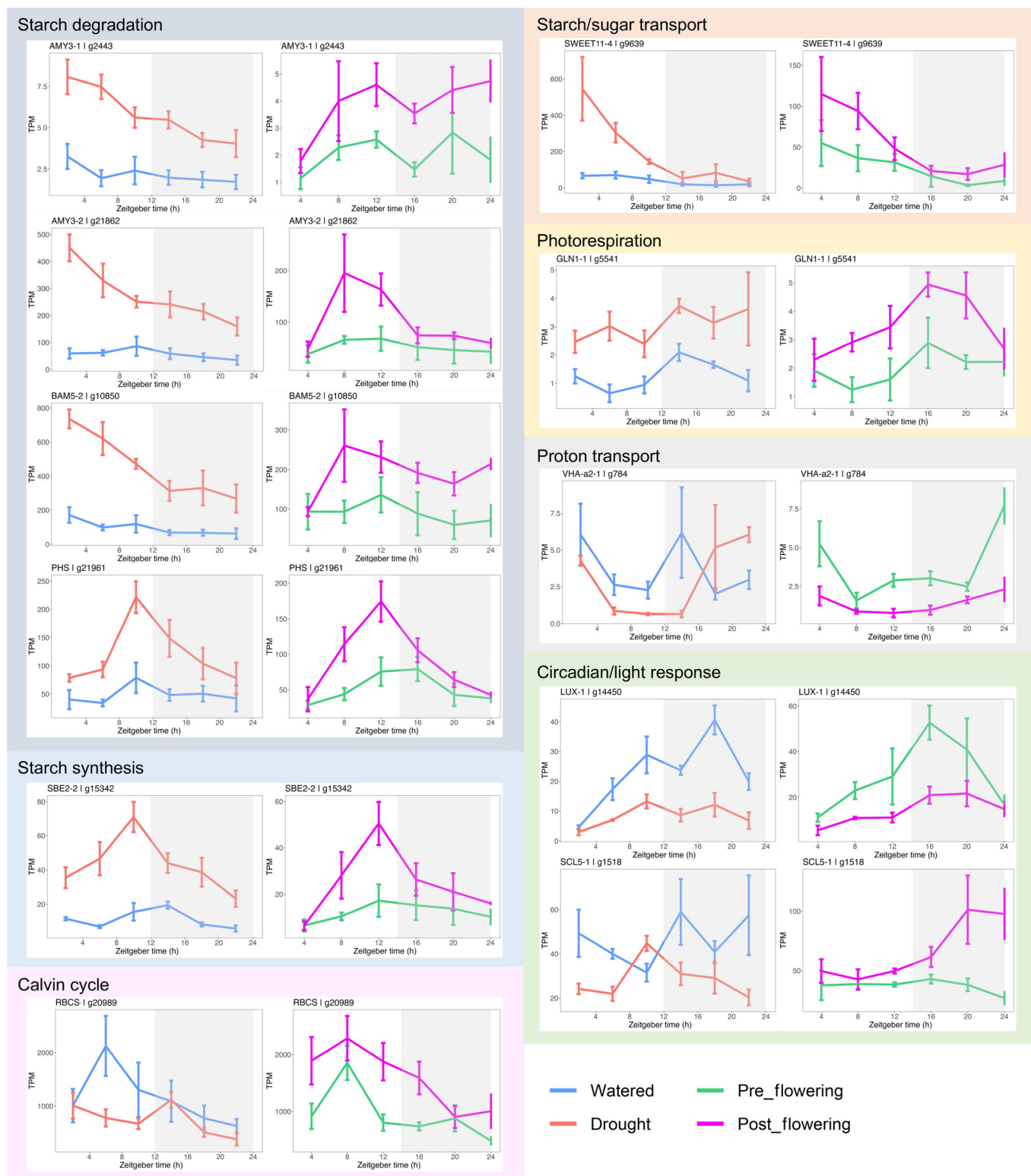


Fig. 5 Expression profiles of 11 photosynthesis pathway genes differentially expressed in both experiments in *Cistanthe cachinalensis*. Normalized abundances of these genes are plotted across time (in transcripts per million), and grouped by photosynthetic pathway identity (as defined by Gilman *et al.*, 2022). In each set of plots, the left panels show the results for the watered vs drought treatment experiment; the right panels show the results for the pre-flowering vs post-flowering stage experiment. Three or four biological replicates for each treatment $\times$ time point or stage $\times$ time point, and error bars indicate SE. AMY, alpha-amylase; BAM, beta-amylase; GLN, glutamine synthetase; LUX, LUX ARRHYTHMO; PHS, alpha-glucan phosphorylase; RBCS, ribulose biphosphate carboxylase (RuBisCO) small subunit; SBE, 1,4-alpha-glucan-branching enzyme; SCL, Scarecrow-like; SWEET, bidirectional sugar transporter; VHA, V-type proton ATPase.

genes–CAM response is more strongly upregulated in drought than in post-flowering plants, but genes we expect to have a diurnal pattern of expression are consistent with a CAM response in the post-flowering plants (Figs 3b,c, 5, S13). Moreover, we found that genes co-expressed with nighttime- (*PPC-1E1c*) and daytime-expressed (*cpNADP-ME4-1*) CAM genes from the post-flowering stage were enriched with regulatory motifs that were unique and absent from their counterparts in the drought treatment (Fig. 4b; Table 2; Dataset S6). This suggests that genes involved in the CAM biochemical module have acquired multiple regulatory motifs that are distinct from those involved in drought response and are shared with genes that are upregulated during reproduction.

Evolution of novel CAM phenotypes via gene regulation and genetic assimilation

Our phenotypic, gene expression, and motif enrichment analyses all tentatively support the emergence of a new kind of C_3 + CAM phenotype, which for this discussion we refer to as ‘reproductive CAM’. It is possible that this phenotype is widespread in plants that already possess a functional CAM cycle, perhaps particularly in annuals that maximally invest in their reproductive output at the end of their lifecycle. We consider the experiments in *C. cachtinalensis* reported here to be preliminary, but we do think they warrant and hopefully will encourage further exploration of CAM expression and reproduction in this and other species. We must also more thoroughly investigate the potential coregulation of CAM and flowering with finer-scale developmental time-series and more powerful molecular tools (e.g. ATAC-seq or DAP-seq; Buenrostro *et al.*, 2013, 2015; Bartlett *et al.*, 2017).

In the meantime, it is worth considering reproductive CAM and its potential evolutionary significance. The simplest explanation might be that, even in well-watered conditions, the simple act of flowering induces some drought stress in individuals, as flowers can often carry significant transpirational burden (Roddy *et al.*, 2023). We did not monitor plant water potentials during the experiment and cannot rule out the possibility that flowering and fruiting plants were more water stressed, even under a similar watering regime as vegetative plants. If this were the case, it would be difficult to consider this a new phenotype, as the plant would simply be upregulating CAM in response to drought stress as it does in other circumstances. Our gene expression data do not really support this model however – there is relatively little overlap in the genes that are co-expressed with CAM genes in the drought and flowering experiments, suggesting that flowering did not simply induce a general ‘drought-stress module’ that happens to include CAM.

Instead, it appears that CAM orthologues have acquired additional regulatory motifs (Fig. 4b; Table 2; Dataset S6) that connect them to genes involved in reproduction, such that CAM and flowering are at least partially coregulated because of a direct recruitment of CAM into a reproductive gene regulatory network/networks. We view this as evolutionarily significant for two main reasons. First, it stands as another example of evolutionary

novelties arising via changes in gene regulation, rather than via adaptive protein origination/evolution (Davidson & Erwin, 2006; Wagner, 2014). With regard to the evolution of CAM specifically, it emphasizes the diversity of phenotypes that can evolve from simple changes in the timing and magnitude of expression of a single biochemical module (Schiller & Bräutigam, 2021; Edwards, 2023).

Second, we speculate that ‘reproductive CAM’ could be an example of evolution via genetic assimilation (Waddington, 1953; West-Eberhard, 1989). The role of stress response in generating novel phenotypes that are eventually genetically assimilated has been receiving fresh attention (‘stress-induced evolutionary innovation’, or SIEI; Wagner *et al.*, 2019; Love & Wagner, 2022). We previously proposed that SIEI could present a useful paradigm for CAM evolution generally – first as an initial assembly of the CAM biochemical module when a plant is out of homeostasis due to stress, and second, as co-option of an established CAM stress response into an obligate strong CAM phenotype (Edwards, 2023). Reproductive CAM would also follow a similar co-option model, whereby the co-expression of CAM and reproduction is at first due to the facultative drought-stress response and the shared timing of flowering and drought in *C. cachtinalensis*’ native environment. But the coordinated timing of CAM induction and flowering is adaptive for other reasons as well, facilitating increased productivity and seed set independently of drought stress, and selection would favor maintaining this coordinated timing of expression. Eventually, this emergent new phenotype could become genetically assimilated via recruitment of the CAM pathway into new gene networks that regulate flowering. The reduced strength of the CAM cycle during flowering as compared to drought, and the mixed expression patterns of CAM genes in the post-flowering stage, suggests that the assimilation and co-option of the full CAM pathway is still incomplete and/or suboptimal in the individuals in our study. It would be well worth surveying other populations of *C. cachtinalensis*, or other *Cistanthe* species in this taxonomically complicated clade, for variation in the strength and regulation of flowering-induced CAM behavior.

Conclusions

We present *C. cachtinalensis* as an excellent new system for understanding the biology and complicated phenotypes of a C_3 + CAM plant. Most C_3 + CAM plants are described as ‘weak constitutive CAM’, showing low levels of CAM expression all the time, or ‘facultative CAM’, where a CAM cycle is induced and reversible (Winter, 2019). But if a plant has a functional CAM cycle, there are other possibilities, at least in theory, for its patterns of expression and its recruitment into other aspects of organismal biology, and these are rarely explored. We show here that the CAM cycle is weakly upregulated when a plant is flowering in *C. cachtinalensis* and suggest that a novel phenotype, ‘reproductive CAM’, may be in the initial stages of formation, perhaps via genetic assimilation of an originally drought-induced co-expression of CAM and flowering (Waddington, 1953; West-Eberhard, 1989; Wagner *et al.*, 2019). Crassulacean acid

metabolism photosynthesis provides great opportunity for studies of complex organismal evolution, yet our working evolutionary model has not matured much beyond the initial speculations of Teeri (1982). Understanding the diversity of phenotypes included in the broad 'C₃ + CAM' umbrella and how they are evolutionarily related to one another and to primary CAM has been an elusive goal but remains central to progress.

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Competing interests

None declared.

Author contributions

AC and EJE designed the project. AC grew and harvested plants for genome sequencing and initiated the assembly, which PR completed, scaffolded and annotated. AC, LW, EGDJ, SGD and VZD grew more plants, performed the experiments and collected data. AC, SGD and VZD analyzed the physiological data, and AC and PR analyzed the RNA-Seq data. AM-M and IEP contributed organismal expertise and interpretation of the data. AC, PR and EJE prepared the manuscript, while all authors edited or reviewed and approved the manuscript.

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Data availability

The genome FASTA file and raw RNA sequencing data are available at NCBI under BioProject PRJNA1181828. The annotation file and scripts for genome assembly are available at

<https://github.com/anriam/Cistanthe-cachinalensis-genome>. A script to process RNA-Seq, raw and normalized RNA-Seq counts, metadata for RNA samples, and R script for DGE analyses and TPM calculations are available at <https://github.com/anriam/Cistanthe-RNASeq>. In the same repository, data and R scripts for visualizing gas exchange and titratable acidity are also available.

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Supporting Information

Additional Supporting Information may be found online in the Supporting Information section at the end of the article.

Dataset S1 RNA-Seq QC statistics.

Dataset S2 List of photosynthetic gene orthologs.

Dataset S3 Outputs from differential gene expression analyses.

Dataset S4 Overlaps in differentially expressed genes.

Dataset S5 List of genes in co-expressed cluster groups.

Dataset S6 Full list of discovered and enriched motifs.

Fig. S1 *Cistanthe cachinalensis* leaf accumulation curves.

Fig. S2 *Cistanthe cachinalensis* chromosome squash image.

Fig. S3 *Cistanthe cachinalensis* Hi-C contact map.

Fig. S4 *Cistanthe cachinalensis* genome scaffolds and core crassulacean acid metabolism gene locations.

Fig. S5 *PEPC1* gene family tree.

Fig. S6 Expression profiles of other core crassulacean acid metabolism pathway genes and gene copies from both experiments.

Fig. S7 *PEPC1* amino acid sequence comparisons among *Cistanthe cachinalensis*, *Portulaca oleraceae*, *Portulaca amilis*, *Kalanchoë fedtschenkoi*, and *Kalanchoë laxiflora*.

Fig. S8 Principal component analysis plots from rlog-transformed normalized gene expression.

Fig. S9 Heatmaps of rlog-transformed normalized gene expression.

Fig. S10 MA plots of mean normalized counts against shrunken log fold change in expression.

Fig. S11 Volcano plots from the 'Drought vs Watered' treatment comparison and 'Post-Flowering vs Pre-Flowering' stage comparison.

Fig. S12 Expression profiles of genes significantly differentially expressed between 'Post-Flowering and Pre-Flowering' stages, but not between 'Drought and Watered' treatments.

Fig. S13 Expression profiles of photosynthetic pathway genes with significant diurnal expression in both 'Drought treatment and Post-Flowering' stage, but not in 'Watered or Pre-Flowering'.

Fig. S14 Clustering of all significant differentially expressed genes in the two experiments via expression profile similarity.

Fig. S15 Clustering of significant differentially expressed photosynthetic pathway genes in the 'Drought vs Watered' module and the distribution of pathway identities.

Fig. S16 Clustering of photosynthetic pathway genes with significant diurnal expression in the 'Drought treatment and Post-Flowering' stage and the distribution of pathway identities.

Fig. S17 Results from Gene Ontology enrichment analyses among the significant differentially expressed genes and gene set enrichment analyses in both experiments.

Fig. S18 Results from Gene Ontology enrichment analyses among genes with significant diurnal expression and gene set enrichment analyses in the 'Drought treatment and the Post-Flowering' stage.

Table S1 *Cistanthe cachinalensis* leaf samples collected across six time points for RNA-Seq and corresponding titratable acidity (ΔH^+) values.

Table S2 Detailed breakdown of repetitive elements of the *Cistanthe cachinalensis* genome.

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